Ex 5: Analysis of Prompting Techniques in LLM Reasoning

Learning Objective:

Develop a Python program to analyze how different prompting strategies—Zero-shot, Few-shot, and Chain-of-Thought (CoT)—affect the reasoning capabilities of Large Language Models (LLMs). The program will also identify patterns in logical fallacies or hallucinations using benchmark datasets.

Steps:

1. Import Required Libraries: OpenAI API or Hugging Face Transformers or Google Gemini API and Additional libraries: datasets, pandas, matplotlib

2. Configure API Keys

- Set your OpenAI and Gemini API keys securely (do not share).

3. Load Dataset

- Use either **SQuAD** (QA reasoning) or **GSM8K** (math reasoning).

4. Initialize Models

5. Define Prompting Strategies

- **Zero-shot Prompt, Few shot prompt, Chain-of-Thought Prompt**

6. Create Functions to Query Models

7. Generate Outputs

- Apply all prompting techniques on the same dataset questions.

8. Display Results of Zero-shot Prompt, Few shot prompt, Chain-of-Thought Prompt

CODE:

```
# Import Libraries
import google.generativeai as genai
from datasets import load_dataset
import pandas as pd
import matplotlib.pyplot as plt
import time
```

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```
genai.configure(api_key="AIzaSyAKzgI4L68do4ufmJe0EiibXt6WnmpT9Ng")
model = genai.GenerativeModel("gemini-2.5-flash")
dataset = load_dataset("gsm8k", "main")
data = dataset["test"].select(range(2))
def ask_gemini(prompt):
    while True:
        try:
            response = model.generate_content(prompt)
            time.sleep(20)
            return response.text
        except Exception as e:
            print("Rate limit reached... waiting 50 sec")
            time.sleep(50)
def zero_shot(q):
    prompt = f"Answer this question:\n{q}"
    return ask_gemini(prompt)
def few_shot(q):
    prompt = f"""
Q: If 2+3 = ?
A: 5
Q: If 10-4 = ?
A: 6

Q: {q}
A:
"""
    return ask_gemini(prompt)
def cot(q):
    prompt = f"Solve step by step:\n{q}"
```

```
    return ask_gemini(prompt)
results = []
for item in data:
    question = item["question"]
    z = zero_shot(question)
    f = few_shot(question)
    c = cot(question)
    results.append({
        "Question": question,
        "Zero-shot": z,
        "Few-shot": f,
        "CoT": c
    })
df = pd.DataFrame(results)
print(df)
techniques = ["Zero-shot", "Few-shot", "CoT"]
lengths = [
    df["Zero-shot"].str.len().mean(),
    df["Few-shot"].str.len().mean(),
    df["CoT"].str.len().mean()
]
plt.bar(techniques, lengths)
plt.title("Prompting Technique Comparison")
plt.xlabel("Technique")
plt.ylabel("Avg Output Length")
plt.show()
```

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OUTPUT:

```
/usr/local/lib/python3.12/dist-packages/google/colab/_import_hooks/_hook_injector.py:55: FutureWarning:

All support for the `google.generativeai` package has ended. It will no longer be receiving
updates or bug fixes. Please switch to the `google.genai` package as soon as possible.
See README for more details:

https://github.com/google-gemini/deprecated-generative-ai-python/blob/main/README.md

  loader.exec_module(module)
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Cola
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and f
README.md: 7.93k/? [00:00<00:00, 112kB/s]
main/train-00000-of-00001.parquet: 100% 2.31M/2.31M [00:00<00:00, 11.5MB/s]
main/test-00000-of-00001.parquet: 100% 419k/419k [00:00<00:00, 1.70MB/s]
```

```
                                                Question  \
0  Janet’s ducks lay 16 eggs per day. She eats th...
1  A robe takes 2 bolts of blue fiber and half th...

                                               Zero-shot  \
0  Here's how to calculate Janet's daily earnings...
1  Here's how to calculate the total bolts:\n\n1....

                                                Few-shot  \
0                                                     18
1  A robe takes 2 bolts of blue fiber and half th...

                                                     CoT
0  Here's how to solve the problem step-by-step:\...
1  Here's how to solve the problem step-by-step:\...
```

Prompting Technique Comparison

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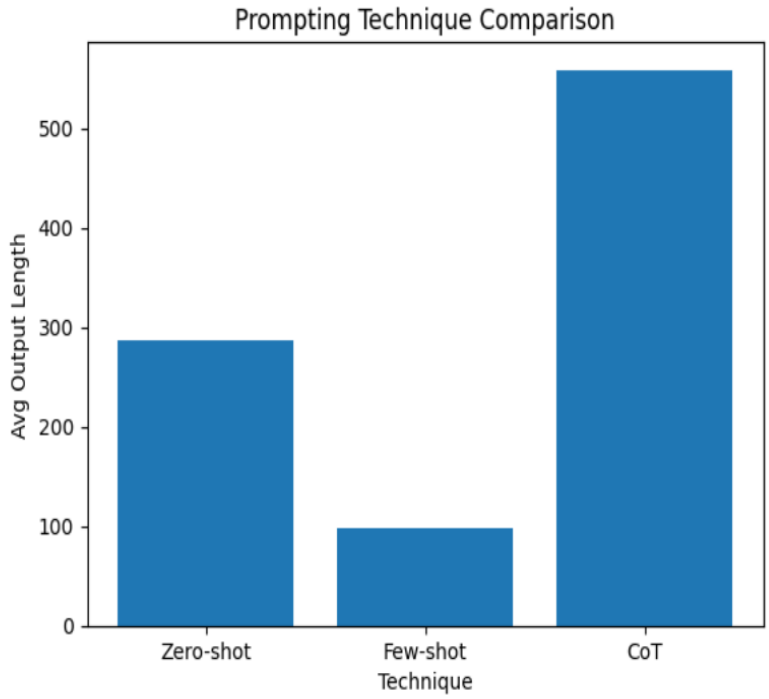

Learning Outcome:

Students will:

- Understand how Zero-shot, Few-shot, and CoT prompting influence LLM reasoning.
- Identify patterns in hallucinations and logical fallacies.
- Gain experience working with benchmark datasets like GSM8K or SQuAD.